

# Chapter 2

# Probability

Probability and Statistics for Science and Engineering  
with Examples in R

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# (1) Sample Spaces and Events

- ▶ Experiment: Any action or process that generates observations
- ▶ Sample Space ( $S$ ): The set of all possible outcomes of the experiment
- ▶ Event: A set of outcomes contained in  $S$

Example 2.1 (continued):

(a) Tossing a coin twice:  $S = \{HH, HT, TH, TT\}$

- ▶ The event of getting exactly one head:  $A = \{HT, TH\}$
- ▶ The event that at least one of the flips results in a head:  
 $B = \{HH, HT, TH\}$

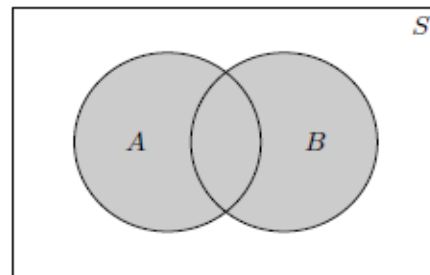
(b) Tossing a die:  $S = \{1, 2, 3, 4, 5, 6\}$

- ▶ The outcome is an odd number:  $\{1, 3, 5\}$
- ▶ The outcome is less than 4:  $\{1, 2, 3\}$
- ▶ The outcome is a 3:  $\{3\}$

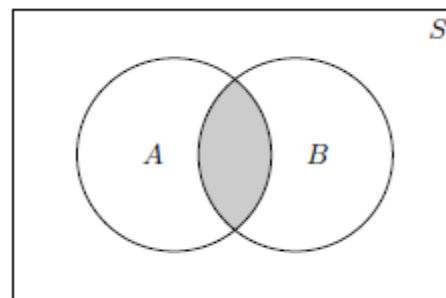
(c) Roll two dice, and record the sum. Then the sample space is

$$S = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

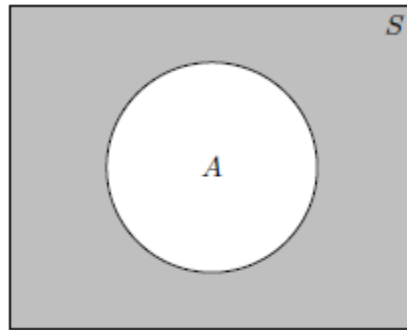
- ▶ Union of two events  $A$  &  $B$  ( $A \cup B$ ;  $A$  or  $B$ ): The event consisting of all outcomes that are either in  $A$  or  $B$  or both



- ▶ Intersection ( $A \cap B$ ;  $A$  and  $B$ ): The event consisting of all outcomes that are in both  $A$  and  $B$



- Complement ( $A^C$ ): not  $A$



### Example 2.2: Tossing a coin twice

$A$ : Tail at the second toss

$B$ : At least one head

$$A = \{HT, TT\}, \quad B = \{HH, HT, TH\}$$

$$A \cup B = \{HH, HT, TH, TT\} = S$$

$$A \cap B = \{HT\}$$

$$A^C = \{HH, TH\}$$

$$B^C = \{TT\}$$

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c$$

(De Morgan's Law)

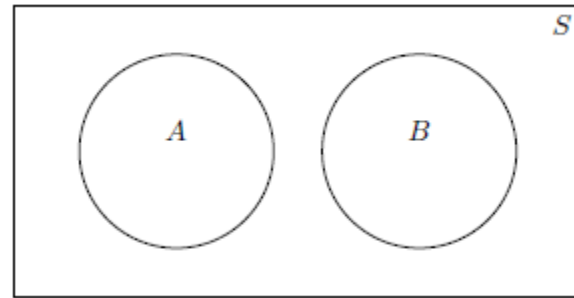
From Example 2.2 (not a proof),

$$(A \cup B)^c = \emptyset = A^c \cap B^c$$

where  $\emptyset$  is the empty set

$$(A \cap B)^c = \{HH, TH, TT\} = A^c \cup B^c$$

- ▶  $A$  and  $B$  are mutually exclusive (disjoint) if  $A$  and  $B$  have no outcomes in common.



- ▶ Example 2.3 (Example 2.2 continued):

$A$ : Tail at the second toss

$D$ : Two heads

$$A = \{HT, TT\}, \quad D = \{HH\}$$

$A$  and  $D$  are disjoint.



## (2) Counting

### A) Tree Diagrams

Example 2.4: Dinner. Choose one from each category:

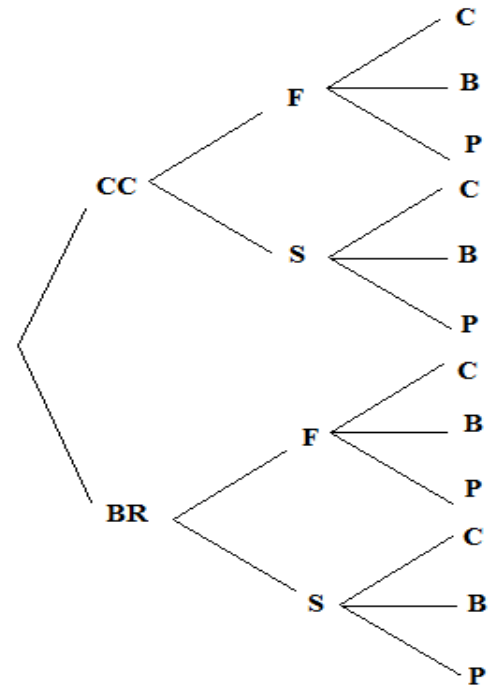
Soup: clam chowder (CC), broccoli (BR)

Side Dish: French fries (F), salad (S)

Main Dish: chicken (C), beef (B), pork (P)

$$2 \times 2 \times 3 = 12$$

possible choices



## B) Product Rule

Suppose a set consists of ordered collections of  $k$  elements.

There are  $n_1$  possible choices for the 1<sup>st</sup> element.

There are  $n_2$  possible choices for the 2<sup>nd</sup> element.

.....

There are  $n_k$  possible choices for the  $k^{\text{th}}$  element.

There are  $n_1 n_2 \cdots n_k$  possible  $k$  combinations.

Example 2.5: If a test consists of 12 true-false questions, in how many different ways can a student mark the test paper with one answer to each question?

$$2 \cdot 2 \cdots 2 = 2^{12} = 4,096$$

Example 2.6: A license plate has exactly 6 characters, each of which can be a digit or a letter excluding letters O and I. (Letters and digits can be repeated.)

(a) How many different license plates are possible?

Let  $A_i = \{0, 1, 2, \dots, 9, A, B, \dots, H, J, \dots, N, P, \dots, Z\}$ ,  
 $i = 1, \dots, 6$

$$\begin{aligned} n_1 \cdots n_6 &= 34 \times 34 \times 34 \times 34 \times 34 \times 34 = 34^6 \\ &= 1,544,804,416 \end{aligned}$$

(b) What if the first 3 characters must be digits, and the last 3 must be letters excluding O and I?

$$\begin{aligned} &\text{number of possible license plates} \\ &= 10 \times 10 \times 10 \times 24 \times 24 \times 24 = 13,824,000 \end{aligned}$$

**Definition** Factorial:

$$n! = n(n - 1)(n - 2) \cdots 2 \cdot 1$$

$$1! = 1$$

$$0! = 1$$

## C) Permutations: Order is important

Definition An ordered sequence of  $k$  objects from a set of  $n$  distinct objects: permutation

$${}_nP_k = n(n-1)(n-2)\cdots(n-k+1) = \frac{n!}{(n-k)!}$$

Example 2.7: A committee consists of 10 members. The number of possible choices of a chair, vice chair, and secretary is

$${}_{10}P_3 = \frac{10!}{(10-3)!} = \frac{10!}{7!} = 10 \cdot 9 \cdot 8 = 720$$

## D) Combinations: Ignore the order

**Definition** Any unordered subset of  $k$  objects from a set of  $n$  distinct objects: combination

$${}_nC_k = \binom{n}{k} = \frac{n(n-1) \cdots (n-k+1)}{k!} = \frac{n!}{k!(n-k)!}$$

**Example 2.8**: A committee consists of 10 members.  
The number of possible choices of 3 representatives is

$$\binom{10}{3} = \frac{10!}{3!(10-3)!} = \frac{10 \cdot 9 \cdot 8}{3!} = \frac{720}{6} = 120$$

Example 2.9: Pick 3 cards in succession from a full deck of 52.

(a) If the order or arrangement of the selection is important, how many possible outcomes?

$${}_{52}P_3 = \frac{52!}{(52-3)!} = \frac{52!}{49!} = 52 \cdot 51 \cdot 50 = 132,600$$

(b) If the order or arrangement of the selection is not important, how many possible outcomes?

$$\binom{52}{3} = \frac{{}_{52}P_3}{3!} = \frac{52!}{3!(52-3)!} = \frac{52 \cdot 51 \cdot 50}{3 \cdot 2} = 22,100$$

### (3) Relative Frequency (Equally Likely Outcomes)

- If there are  $n$  equally likely outcomes, and  $s$  are labeled success, then the probability of a successful outcome is given as  $s/n$ .

Example 2.10: Tossing a fair die:  $S = \{1, 2, 3, 4, 5, 6\}$

Probability of getting an odd number  $A = \{1, 3, 5\}$  is

$$P(A) = \frac{\text{\#elements in } A}{\text{\#elements in } S} = \frac{3}{6} = \frac{1}{2}.$$



Example 2.11: Probability of drawing a king from a deck of 52 cards:

Let  $A = \{\text{king}\}$ . Then

$$P(A) = \frac{\text{\#elements in } A}{\text{\#elements in } S} = \frac{4}{52} = \frac{1}{13}$$

Example 2.12: Tossing a fair die twice

Let  $A = \{\text{sum of the numbers is 6}\}$   
 $= \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$ .

The probability of  $A$  is

$$P(A) = \frac{\text{\#elements in } A}{\text{\#elements in } S} = \frac{5}{36}$$

## (4) Probability

### ► Probability Axioms

1.  $0 \leq P(A) \leq 1$
2.  $P(S) = 1$
3. If  $A$  and  $B$  are any mutually exclusive events in  $S$ , then
$$P(A \cup B) = P(A) + P(B)$$

### ► Generalization of 3:

If  $A_1, A_2, \dots, A_n$  are mutually exclusive events in  $S$ , then
$$P(A_1 \cup A_2 \cup \dots \cup A_n) = \sum_{i=1}^n P(A_i).$$

Example 2.14: Tossing a fair coin:  $S = \{H, T\}$

$$P(H) = P(T) = \frac{1}{2}$$
$$P(S) = P(H) + P(T)$$

Example 2.15: Tossing two balanced coins:

$$S = \{HH, HT, TH, TT\}$$

$$P(S) = P(HH) + P(HT) + P(TH) + P(TT) = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = 1$$

Let  $A$  be the event of obtaining at least one tail. Then

$$A = \{HT, TH, TT\}$$

$$P(A) = P(HT) + P(TH) + P(TT) = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}$$

Example 2.16: Tossing a fair coin until a tail is obtained

$$S = \{T, HT, HHT, \dots\}$$

$$\begin{aligned} P(S) &= P(T) + P(HT) + P(HHT) + \dots \\ &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n} + \dots = \frac{\frac{1}{2}}{1 - \frac{1}{2}} = 1 \end{aligned}$$

- Law of Complementation:  $P(A^C) = 1 - P(A)$

Proof)

$$S = A \cup A^C$$

$$1 = P(S) = P(A \cup A^C) = P(A) + P(A^C)$$

Thus,  $P(A^C) = 1 - P(A)$ .

- If  $A \subset B$ , then  $P(A) \leq P(B)$ .

Example 2.17: Tossing a fair coin twice

$$S = \{HH, HT, TH, TT\}$$

Let  $A = \{\text{two heads}\} = \{HH\}$ . Then

$$A^C = \{\text{at least one is tail}\} = \{HT, TH, TT\}.$$

$$P(A) = \frac{1}{4}, P(A^C) = \frac{3}{4}$$

$$P(A^C) = 1 - P(A)$$

►  $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Proof)

$$\begin{aligned} P(A \cup B) &= P(A \cap B) + P(A \cap B^c) + P(A^c \cap B) \\ &= [P(A \cap B) + P(A \cap B^c)] + [P(A \cap B) + P(A^c \cap B)] - P(A \cap B) \\ &= P(A) + P(B) - P(A \cap B) \end{aligned}$$

► If  $A$  and  $B$  are mutually exclusive, then

$$P(A \cup B) = P(A) + P(B)$$

Example 2.18: Let 80% of freshmen in a college take statistics, 50% take physics and 40% take both statistics and physics.

(a) What is the probability of taking at least one of these courses?

Let  $A = \{\text{taking statistics}\}$ ,  $B = \{\text{taking physics}\}$ . Then

$$P(A) = 0.8, P(B) = 0.5, P(A \cap B) = 0.4$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.8 + 0.5 - 0.4 = 0.9$$

(b) What is the probability of taking only one of these courses?

$$P(A \cup B) - P(A \cap B) = 0.9 - 0.4 = 0.5$$

► 
$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$



Example 2.19: A committee has 8 male members and 12 female members. Choose 5 representatives in this committee at random. Let  $D_i = \{\text{exactly } i \text{ of the 5 representatives are females}\}$ ,  $i = 0, 1, \dots, 5$ .

(a) Probability that exactly 3 of the 5 representatives are females?

$$P(D_3) = \frac{\binom{12}{3} \binom{8}{2}}{\binom{20}{5}} = \frac{\frac{12!}{3!9!} \cdot \frac{8!}{2!6!}}{\frac{20!}{5!15!}} = 0.3973$$

(b) Probability that at least 3 of the 5 representatives are females?

$$P(D_3 \cup D_4 \cup D_5) = \frac{\binom{12}{3} \binom{8}{2}}{\binom{20}{5}} + \frac{\binom{12}{4} \binom{8}{1}}{\binom{20}{5}} + \frac{\binom{12}{5} \binom{8}{0}}{\binom{20}{5}} = 0.7038$$

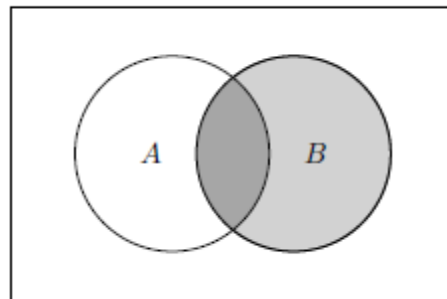
## (5) Conditional Probability

- Conditional probability of  $A$  given  $B$ :

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \text{ if } P(B) > 0$$

Equivalently,

$$\begin{aligned} P(A \cap B) &= P(B)P(A|B) \text{ if } P(B) > 0 \\ &= P(A)P(B|A) \text{ if } P(A) > 0 \end{aligned}$$



### Example 2.21: BMI and age group

|          | Normal<br>or Low<br>BMI | Overwei<br>ght | Obese | Total |
|----------|-------------------------|----------------|-------|-------|
| Age < 30 | 0.09                    | 0.06           | 0.05  | 0.20  |
| Age ≥ 30 | 0.20                    | 0.32           | 0.28  | 0.80  |
| Total    | 0.29                    | 0.38           | 0.33  | 1.00  |

(a) What is the probability that a person selected at random from the group will be obese?

Let  $A = \{\text{obese}\}$ . Then  $P(A) = 0.33$ .

(b) A person, selected at random from this group, is found to be obese. What is the probability that this person is younger than 30 years old?

Let  $B = \{\text{younger than 30 years old}\}$ . Then

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{0.05}{0.33} = 0.15$$

► Conditional probability: Probability

$$P(A^c|B) = 1 - P(A|B)$$

## (6) Independence

- $A$  and  $B$  are independent if  $P(A|B) = P(A)$ .

### Example 2.22: Tossing a fair die

Let  $A = \{2, 4, 6\}$ ,  $B = \{1, 2, 3\}$ ,  $C = \{1, 2, 3, 4\}$ .

Then  $P(A) = \frac{1}{2}$ ,  $P(A|B) = \frac{1}{3}$ ,  $P(A|C) = \frac{1}{2}$ .

$A$  and  $B$  are dependent &  $A$  and  $C$  are independent.

- $A$  and  $B$  are independent

$$\Leftrightarrow P(A \cap B) = P(A)P(B),$$

$$P(A \cap B^c) = P(A)P(B^c), \quad P(A^c \cap B^c) = P(A^c)P(B^c)$$

Proof)  $P(A \cap B) = P(A|B)P(B)$

and by independence  $P(A|B) = P(A)$

Therefore,  $P(A \cap B) = P(A)P(B)$

Example 2.22 (continued)

$$A = \{2, 4, 6\}, \quad B = \{1, 2, 3\}, \quad C = \{1, 2, 3, 4\}$$

$A \cap B = \{2\}$ .  $A$  and  $B$  are dependent because

$$P(A \cap B) = \frac{1}{6} \neq P(A)P(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$A \cap C = \{2, 4\}$ .  $A$  and  $C$  are independent because

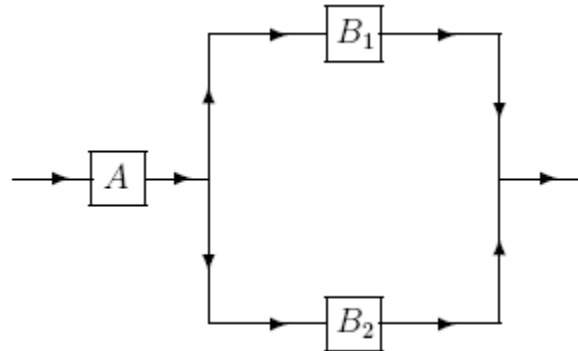
$$P(A \cap C) = \frac{1}{3} = P(A)P(C) = \frac{1}{2} \cdot \frac{2}{3}$$

- ▶  $A_1, A_2, \dots, A_n$  are independent if, for any  $k$ , and every subset of indices  $i_1, i_2, \dots, i_k$ ,  
$$P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}) = P(A_{i_1})P(A_{i_2}) \cdots P(A_{i_k}).$$

### Example 2.23:

$$P(A) = 0.9, P(B_1) = 0.8 \text{ and } P(B_2) = 0.7$$

$A$ ,  $B_1$  and  $B_2$  operate independently. The following system works if  $A$  works and either  $B_1$  or  $B_2$  works. What is the probability that the following system works?



$$\begin{aligned} P(\text{system working}) &= P[A \cap (B_1 \cup B_2)] \\ &= P(A)P(B_1 \cup B_2) \quad (\text{by independence}) \\ &= 0.9[P(B_1) + P(B_2) - P(B_1 \cap B_2)] \\ &= 0.9[P(B_1) + P(B_2) - P(B_1)P(B_2)] \\ &= 0.9[0.8 + 0.7 - (0.8)(0.7)] \\ &= 0.9(0.94) = 0.846 \end{aligned}$$

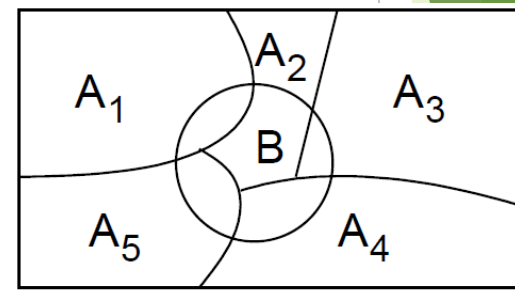


# (7) Bayes' Theorem

- $A_1, A_2, \dots, A_n$  are mutually exclusive if  $A_i \cap A_j = \phi$  for all  $i$  &  $j$ .
- $A_1, A_2, \dots, A_n$  are exhaustive if  $A_1 \cup \dots \cup A_n = S$ .

If  $A_1, A_2, \dots, A_n$  are mutually exclusive & exhaustive, then

$$\begin{aligned} P(B) &= P(B \cap A_1) + \dots + P(B \cap A_n) \\ &= P(A_1)P(B|A_1) + \dots + P(A_n)P(B|A_n) \\ &= \sum_{i=1}^n P(A_i)P(B|A_i) \end{aligned}$$



## Bayes' Theorem:

Let  $A_1, A_2, \dots, A_n$  be mutually exclusive & exhaustive. Then

$$P(A_r|B) = \frac{P(A_r \cap B)}{P(B)} = \frac{P(A_r)P(B|A_r)}{\sum_{i=1}^n P(A_i)P(B|A_i)}, r = 1, 2, \dots, n$$

Example 2.24: Probability of HIV infection in the US: 0.4%

Let  $A_1 = \{\text{individual has HIV}\}$

$A_2 = \{\text{individual does not have HIV}\}$

$B = \{\text{positive HIV test result}\}$

$$P(A_1) = 0.004; P(A_2) = 0.996$$

$$P(B|A_1) = 0.997; P(B|A_2) = 0.015$$

If a test result is positive, what is the probability that the individual has the disease?

$$\begin{aligned} P(A_1|B) &= \frac{P(A_1 \cap B)}{P(B)} = \frac{P(A_1)P(B|A_1)}{\sum_{i=1}^2 P(A_i)P(B|A_i)} \\ &= \frac{(0.004)(0.997)}{(0.004)(0.997) + (0.996)(0.015)} = 0.21 \end{aligned}$$